<Xpro-Cancer Prediction Project>

Software Requirements Specification

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Table of Contents

1. Introduction

1.1 Purpose
1.2 Intended Audience and Reading Suggestions
1.3 Project Scope

2. Overall Description

2.1 Project Perspective
2.2 Project Functions
2.3 User Classes and Characteristics
2.4 Operating Environment
2.5 Design and Implementation Constraints
2.6 User Documentation
2.7 Assumptions and Dependencies

3. System Features

3.1 Data Preparation
3.2 Data Cleaning
3.3 Model Training
3.4 Model Evaluation & Explainability
3.5 Prediction Output

4. Other Nonfunctional Requirements

4.1 Performance Requirements
4.2 Safety Requirements
4.3 Security Requirements
4.4 Software Quality Attributes

5. Risks and Mitigation

5.1 Risks Table
5.1 Mitigation Plan

6. Screenshots

APPENDIX

1. Introduction

1.1 Purpose

The purpose of this project is to design a machine learning–based system that predicts the likelihood of cancer using synthetic patient data. It provides students with hands-on experience in the complete AI/ML lifecycle while ensuring data privacy and ethical practices. Key objectives include:

- Building a pipeline for data generation, anomaly handling, and cleaning.
- Training and evaluating multiple ML models with standard metrics.
- Producing explainable predictions using feature importance and SHAP values.
- Developing professional documentation (SRS, PM plan, final report) to simulate real-world project delivery.

In essence, the project's purpose is to combine technical implementation with professional skills, preparing students for real-world AI applications in healthcare.

1.2 Intended Audience and Reading Suggestions

• **Students/Interns**→ Understand dataset creation, anomaly handling, and ML model design.

• **Faculty/Mentors** → Evaluate methodology, progress, and deliverables.

• **Industry Evaluators** → Assess project planning, documentation, and system robustness.

1.3 Product Scope

The Cancer Prediction project simulates a real-world AI/ML pipeline using synthetic clinical data. The scope defines what is included and excluded in this internship project.

In Scope

- **Data Generation**: Create ≥1000 synthetic patient records with demographic, lifestyle, biomarker, and symptom attributes.
- **Anomaly Injection**: Introduce realistic errors (null values, duplicates, outliers, incorrect formats) to mimic real-world datasets.
- **Data Cleaning & Validation**: Implement preprocessing techniques (imputation, duplicate removal, outlier detection, format correction) and verify dataset readiness.
- **Machine Learning Model Development**: Train and evaluate multiple ML models (Logistic Regression, Random Forest, XGBoost) for cancer prediction.

- **Performance Evaluation**: Measure models using metrics such as Accuracy, Precision, Recall, F1, and AUROC.
- **Explainability**: Provide feature importance and SHAP-based explanations to ensure transparency.
- **Documentation & Project Management**: Deliver SRS, Project Management Plan, and final project report following professional standards.
- **Educational Simulation**: Serve as a learning-focused exercise to enhance AI/ML and project management skills.

Out of Scope

- Use of real patient or sensitive medical data.
- Integration with hospital systems (EHR, PACS).
- Clinical validation, regulatory compliance, or deployment in live environments.
- Advanced deep learning methods beyond internship requirements.

2. Overall Description

NOTE:
This section contains the general view of the project and the environment where it will be used, describes the expected user audience and constraints, assumptions and dependencies which can be identified.

2.1 Product Perspective

The Cancer Prediction project is a **standalone AI/ML prototype** developed in an academic internship setting. It simulates a healthcare AI workflow using **synthetic patient data** to ensure privacy and ethical compliance.

The system is structured as a modular pipeline:

1. Data Generation & Preprocessing – create patient records, inject anomalies, and clean the dataset.

2. Model Development & Evaluation – train machine learning models and assess performance.

3. Explainability & Output – provide risk predictions with feature importance for transparency.

While independent, the design can be extended in the future to integrate with hospital information systems or clinical validation environments.

2.2 Product Functions

• Synthetic Data Generation → Create ≥1000 patient records with cancer risk factors.

• Data Cleaning & Validation → Handle missing values, outliers, duplicates, and errors.

• ML Model Development → Train and evaluate Logistic Regression, Random Forest, and XGBoost.

• Performance Evaluation → Use Accuracy, Precision, Recall, F1-score, and AUROC with visualizations.

• Explainability & Prediction Output → Provide risk predictions with feature importance and probability scores.

2.3 User Classes and Characteristics

- Interns/Students**:** Implement and test; beginner to intermediate coding skills.
- Mentors/Faculty: Review deliverables; academic knowledge in AI.
- Industry Evaluators**:** Judge based on practicality, completeness, and professionalism.
-

2.4 Operating Environment

- Software: Windows/Linux, Python 3.10+, Jupyter Notebook, Excel.
- Hardware: Standard laptop (8 GB RAM, Intel i5). GPU optional.
- Tools**:** Google Colab, GitHub for collaboration.

2.5 Design and Implementation Constraints

- Limited to 6 weeks.
- No sensitive patient data allowed.
- Reproducibility ensured via random seeds.
- Open-source libraries only.

2.6 User Documentation

- SRS Document (this file).
- Project Plan (Excel).
- Final Technical Report.
- README.md with installation & execution steps.

2.7 Assumptions and Dependencies

- Synthetic dataset simulates real-world patterns.
- Balanced dataset assumed, else handled with resampling.
- Dependencies: NumPy, Pandas, scikit-learn, Matplotlib.

3. System Features

3.1 Data Preparation

- The system shall generate at least **1000 synthetic patient records** with demographic, clinical, and biomarker attributes.
- The system shall inject anomalies such as **missing values, duplicates, and outliers** to simulate real-world datasets.

3.2 Data Cleaning

- The system shall detect and remove duplicate records.
- The system shall handle missing values using **mean/median/mode imputation**.
- The system shall identify and treat outliers using **IQR and Z-score methods**.
- The system shall ensure correct formatting for all attributes.

3.3 Model Training

- The system shall train at least **three ML models**: Logistic Regression, Random Forest, and XGBoost.
- The system shall allow **cross-validation and hyperparameter tuning** for better performance.

3.4 Model Evaluation & Explainability

- The system shall evaluate models using metrics such as **Accuracy, Precision, Recall, F1-score, and AUROC**.

- The system shall generate **confusion matrices, ROC curves, and PR curves** for performance visualization.
- The system shall provide **global feature importance** and **local explanations (SHAP values)** for interpretability.

3.5 Prediction Output

- The system shall accept new patient records in **CSV/Excel format** as input.
- The system shall output a **predicted cancer risk (Yes/No)** with an associated **probability score**.
- The system shall generate an **output report file** summarizing predictions.

4. Other Nonfunctional Requirements

4.1 Performance Requirements

- Training time < 5 minutes for 1000 records.
- Inference latency < 1 second per record.

4.2 Safety Requirements

- Synthetic dataset ensures no sensitive info.
- For research/learning only, not clinical use.

4.3 Security Requirements

- Code stored in private GitHub repo.
- Data files restricted to team only.

4.4 Software Quality Attributes

- Usability: Well-documented Jupyter notebook.
- Reliability: Consistent predictions on test runs.
- Maintainability: Modular, PEP8-compliant code.
- Portability: Works on Windows/Linux, Colab.

5. Risks and mitigation

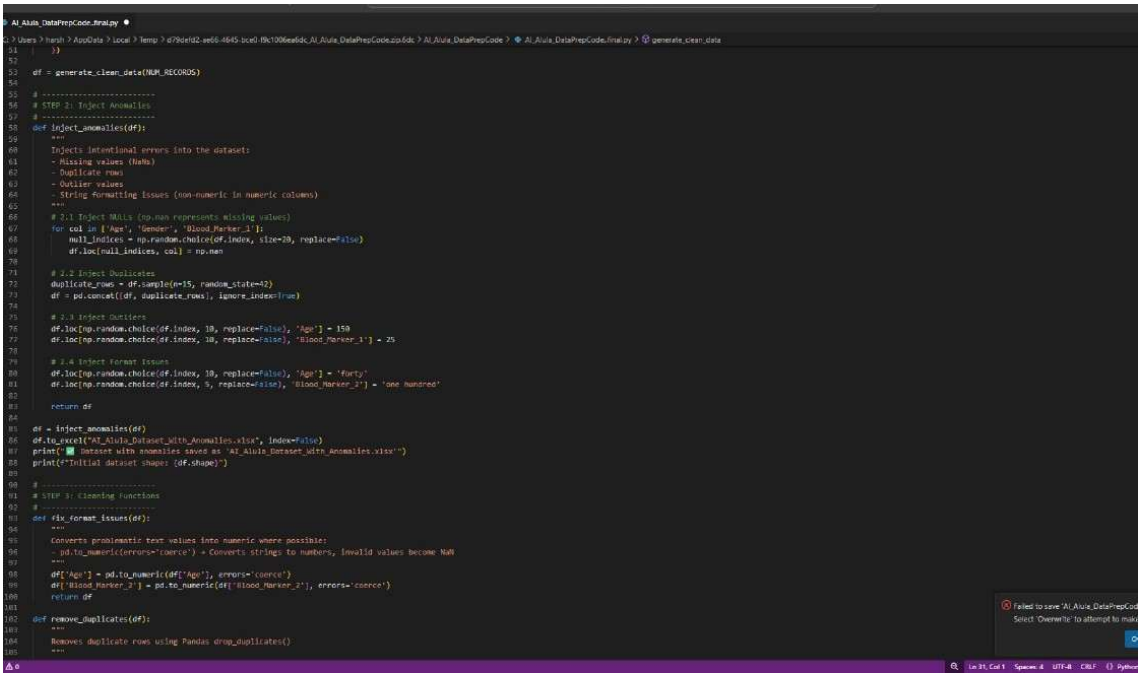
6.1 Risks Table

ID	Description	Likelihood	Impact	Status
R1	Dataset imbalance may reduce recall for cancer	Medium	High	Open
R2	Data cleaning may remove useful records	Medium	Medium	Open
R3	Limited compute power for ML training	Low	Medium	Open
R4	Timeline delays due to multiple deliverables	Medium	High	Open
R5	Dependency issues with Python libraries	Low	Medium	Open

6.2 Mitigation Table

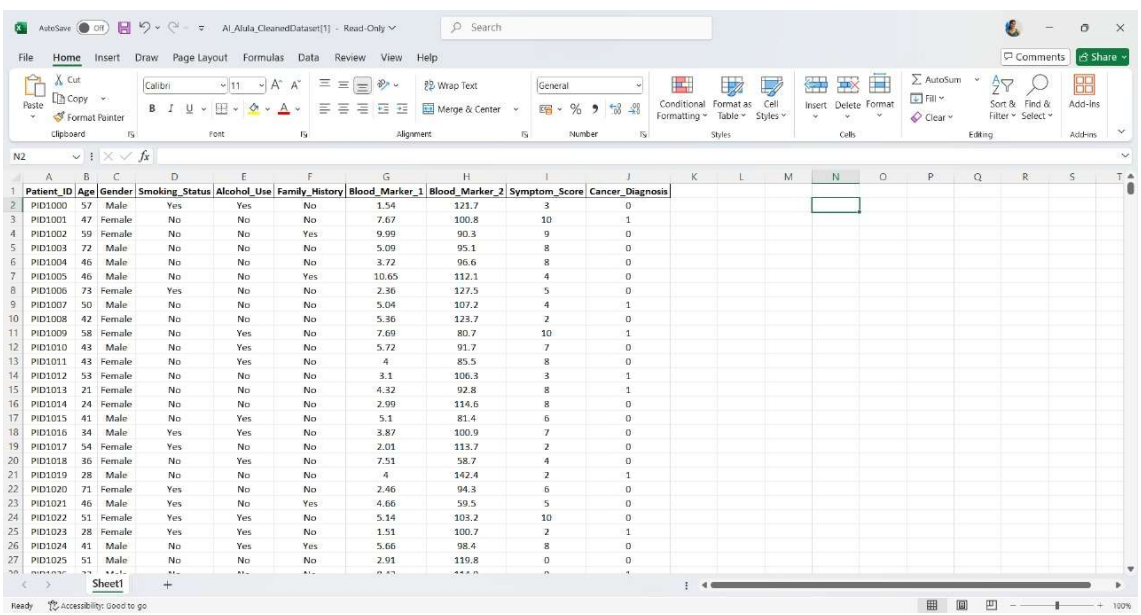
Risk/Issue ID	Mitigation Strategy	Monitoring Method
R1/I1	Use SMOTE or class weighting for imbalance	Compare recall across classes
R2/I2	Careful thresholds for anomaly removal	Run validation script
R3	Use lightweight models first (LogReg, RF)	Check runtime per model
R4	Follow PM plan strictly, track weekly	Weekly progress check
R5	Fix versions in requirements.txt	Test environment setup

6. Screenshots



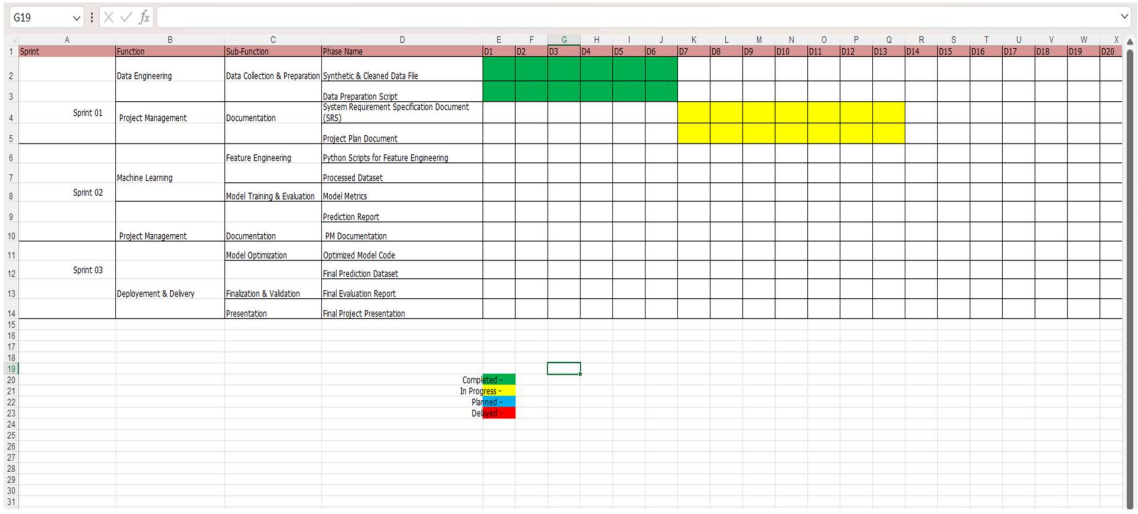
```python
df = generate_clean_data(NUM_RECORDS)

# ------------------------
# STEP 2: Inject Anomalies
# ------------------------
def inject_anomalies(df):
    """
    Injects intentional errors into the dataset:
    - Missing values (NaNs)
    - Duplicate rows
    - Outlier values
    - String formatting issues (non-numeric in numeric columns)
    """
    # 2.1 Inject NAN's (np.nan represents missing values)
    for col in ['Age', 'Gender', 'Blood_Marker_1']:
        null_indices = np.random.choice(df.index, size=20, replace=False)
        df.loc[null_indices, col] = np.nan

    # 2.2 Inject Duplicates
    duplicate_rows = df.sample(n=15, random_state=42)
    df = pd.concat([df, duplicate_rows], ignore_index=True)

    # 2.3 Inject Outliers
    df.loc[np.random.choice(df.index, 10, replace=False), 'Age'] = 150
    df.loc[np.random.choice(df.index, 10, replace=False), 'Blood_Marker_1'] = 25

    # 2.4 Inject Format Issues
    df.loc[np.random.choice(df.index, 10, replace=False), 'Age'] = 'Forty'
    df.loc[np.random.choice(df.index, 5, replace=False), 'Blood_Marker_2'] = 'one hundred'

    return df

df = inject_anomalies(df)
df.to_excel("AI_Alula_Dataset_With_Anomalies.xlsx", index=False)
print("✅ Dataset with anomalies saved as 'AI_Alula_Dataset_With_Anomalies.xlsx'")
print(f"Initial dataset shape: {df.shape}")

# ------------------------
# STEP 3: Cleaning Functions
# ------------------------
def fix_format_issues(df):
    """
    Converts problematic text values into numeric where possible:
    - pd.to_numeric(errors='coerce') + Converts strings to numbers, invalid values become NaN
    """
    df['Age'] = pd.to_numeric(df['Age'], errors='coerce')
    df['Blood_Marker_2'] = pd.to_numeric(df['Blood_Marker_2'], errors='coerce')
    return df

def remove_duplicates(df):
    """
    Removes duplicate rows using Pandas drop_duplicates()
    """
```

Inject Anomalies



Patient_ID	Age	Gender	Smoking_Status	Alcohol_Use	Family_History	Blood_Marker_1	Blood_Marker_2	Symptom_Score	Cancer_Diagnosis
PID1000	57	Male	Yes	Yes	No	1.54	121.7	3	0
PID1001	47	Female	No	No	No	7.67	100.8	10	1
PID1002	59	Female	No	No	Yes	9.99	90.3	9	0
PID1003	72	Male	No	No	No	5.09	95.1	8	0
PID1004	46	Male	No	No	No	3.72	96.6	8	0
PID1005	46	Male	No	No	Yes	10.65	112.1	4	0
PID1006	73	Female	Yes	No	No	2.36	127.5	5	0
PID1007	50	Male	No	No	No	5.04	107.2	4	1
PID1008	42	Female	No	No	No	5.36	123.7	2	0
PID1009	58	Female	No	Yes	No	7.69	80.7	10	1
PID1010	43	Male	No	Yes	No	5.72	91.7	7	0
PID1011	43	Female	No	Yes	No	4	85.5	8	0
PID1012	53	Female	No	No	No	3.1	106.3	3	1
PID1013	21	Female	No	No	No	4.32	92.8	8	1
PID1014	24	Female	No	No	No	2.99	114.6	8	0
PID1015	41	Male	No	Yes	No	5.1	81.4	6	0
PID1016	34	Male	Yes	Yes	No	3.87	100.9	7	0
PID1017	54	Female	Yes	No	No	2.01	113.7	2	0
PID1018	36	Female	No	Yes	No	7.51	58.7	4	0
PID1019	28	Male	No	No	No	4	142.4	2	1
PID1020	71	Female	Yes	No	No	2.46	94.3	6	0
PID1021	46	Male	Yes	No	Yes	4.66	59.5	5	0
PID1022	51	Female	Yes	Yes	No	5.14	103.2	10	0
PID1023	28	Female	Yes	Yes	No	1.51	100.7	2	1
PID1024	41	Male	No	Yes	Yes	5.66	98.4	8	0
PID1025	51	Male	No	No	No	2.91	119.8	0	0

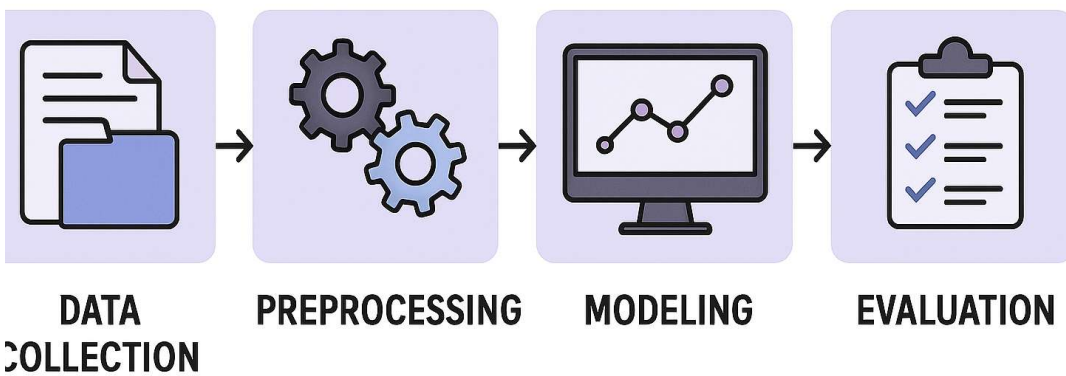
Cleaned Dataset



Sprint	Function	Sub-Function	Phase Name	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
Sprint 01	Data Engineering	Data Collection & Preparation	Synthetic & Cleaned Data File																				
			Data Preparation Script																				
	Project Management	Documentation	System Requirement Specification Document (SRS)																				
			Project Plan Document																				
Sprint 02	Machine Learning	Feature Engineering	Python Scripts for Feature Engineering																				
			Processed Dataset																				
		Model Training & Evaluation	Model Metrics																				
			Prediction Report																				
	Project Management	Documentation	PM Documentation																				
Sprint 03	Deployement & Delivery	Model Optimization	Optimized Model Code																				
			Final Prediction Dataset																				
		Finalization & Validation	Final Evaluation Report																				
		Presentation	Final Project Presentation																				

Completed -
In Progress -
Planned -
Delayed -

Project Plan



Project Workflow

Appendix

6.1 Libraries Used

The following Python libraries were used by **Team Alula** for data generation, anomaly injection, cleaning, and export:

- **faker** → Generate synthetic patient demographics and attributes.
- **pandas** → Data handling, manipulation, and Excel export.
- **NumPy** → Numerical computations and randomization.
- **SciPy** → Statistical analysis and Z-score based anomaly detection.
- **openpyxl** → Excel file read/write operations.
- **random** → Randomization for value selection and anomaly creation.
- **datetime** → Handling diagnosis and follow-up dates.

6.2 Output Files

- **patient_data.xlsx** → Raw dataset with 1000 patient records (includes anomalies).
- **AI_Alula_CleanedDataset.xlsx** → Cleaned dataset after duplicate removal, null handling, and outlier treatment.

6.3 Prototype Implementation (Code Appendix)

A. Data Generation

- Used **Faker** for personal details (names, jobs, addresses).
- Used **Random** for clinical values in logical ranges.
    - Tumor Size (0.5–10 cm), Hemoglobin (8–17 g/dL), Age (20–90 years).
- Created patient records with **demographics, clinical features, treatment & outcome details**.
- Ensured valid medical dates (Diagnosis Date 2010–2024, Last Visit ≥ Diagnosis Date).
- Exported raw dataset → patient_data.xlsx.

B. Introducing Anomalies

- **Outliers (7%)** → Extreme values injected randomly into numerical columns.
- **Null Values (10%)** → Random missing cells across dataset.
- **Duplicates (30%)** → Random row duplication + shuffle.

C. Data Cleaning

- Removed duplicates, replaced missing values (median/mode), corrected dates.
- Detected & removed outliers (Z-score > 3).
- Reassigned sequential **Patient IDs** for consistency.
- Exported clean dataset → AI_Alula_CleanedDataset.xlsx.